

STRATOSCOPE / ATMOSPHERIC PULSECHECK

NORTHERN HEMISPHERE • 2026–27 STRATOSPHERIC SEASON

ISSUED 09 OCT 2026 | UTC

01 LATEST STATE / ENSEMBLES

10 hPa, 60°N zonal-mean u | Valid 09 Oct 00 UTC

GEFS 08 Oct 18Z +15.4 m/s (31 members)
IFS ENS 08 Oct 12Z +15.1 m/s (51 members)
AIFS ENS 08 Oct 18Z +15.2 m/s (51 members)

0 / 133 easterly members through respective +15–16d ranges

02 RUN EVOLUTION / 07–08 OCT

Same valid time: 18 Oct 00 UTC

GEFS +22.2 → +20.6 m/s (−1.7)
IFS ENS +17.3 → +17.9 m/s (+0.6)
AIFS ENS +20.4 → +20.5 m/s (+0.2)

Available cycle history ≈30 h; not a complete 7-day audit

03 EMERGING SIGNALS / LIMITS

- 10 hPa polar-cap ensemble means cool through 18 Oct.
- GEFS 100 hPa eddy heat flux reaches 9.9 K m/s on 11 Oct, falling to 2.8 K m/s by 16 Oct.
- Oct-issued ECMWF seasonal mean remains westerly through December; extended confidence is low.
- AO / NAO / MJO and downward coupling not verified.

Heat-flux values are not anomalies; ERA5 1993–2016 is a baseline only.

04 DECISION BRIEF / LOOK-AHEAD

VORTEX CONSOLIDATION; NO NEAR-TERM REVERSAL

0–72 h	Westerlies persist; modest increase	HIGH
Day 4–10	Strengthening and cooling	HIGH
Day 11–16	Westerlies persist; spread grows	MODERATE

Caveat: deterministic wind fields not independently extracted; tropospheric causality is unsupported.

SOURCE: Stratoscope published northern diagnostics, forecast-data-diagnostics (GEFS / IFS ENS / AIFS ENS), 08 Oct cycles; GEFS vortex heat-flux; ECMWF seasonal Oct issue.

Diagnostic briefing only • Not a substitute for official forecast products • Forecast data are not observations

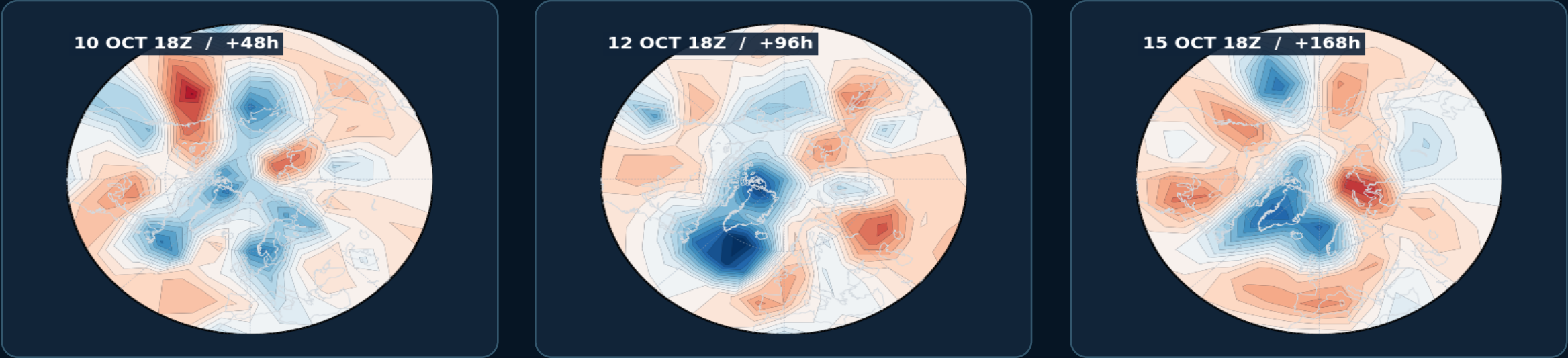
STRATOSCOPE / H500 ANOMALY EVOLUTION

GEFS 31-member mean | 08 OCT 18Z initialisation | 500 hPa height - NCEP/NCAR R1 1991-2020 daily climatology

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FORECAST EVOLUTION, NOT AN OBSERVED 7-DAY RETROSPECTIVE

Blue: negative | red: positive | ±300 m | 10° sample grid



H500 REGIONAL HEIGHT-ANOMALY SIGNAL / AREA-WEIGHTED 45-75°N

VALID (UTC)	N ATL 60°W-0°	EUROPE 0-60°E	ASIA 60-150°E	N PAC 150-240°E	N AMERICA 120-60°W	POLAR 75-90°N
10 Oct 18Z	-52 m	-66 m	+30 m	+3 m	+2 m	-52 m
12 Oct 18Z	-160 m	+58 m	+49 m	-23 m	-5 m	-80 m
15 Oct 18Z	-91 m	+16 m	+29 m	+26 m	+26 m	-22 m
19 Oct 06Z	-63 m	+36 m	+37 m	+18 m	+5 m	-16 m
22 Oct 18Z	-30 m	+36 m	+42 m	+19 m	-27 m	+18 m

RECENT HISTORY / IMPORTANT LIMIT

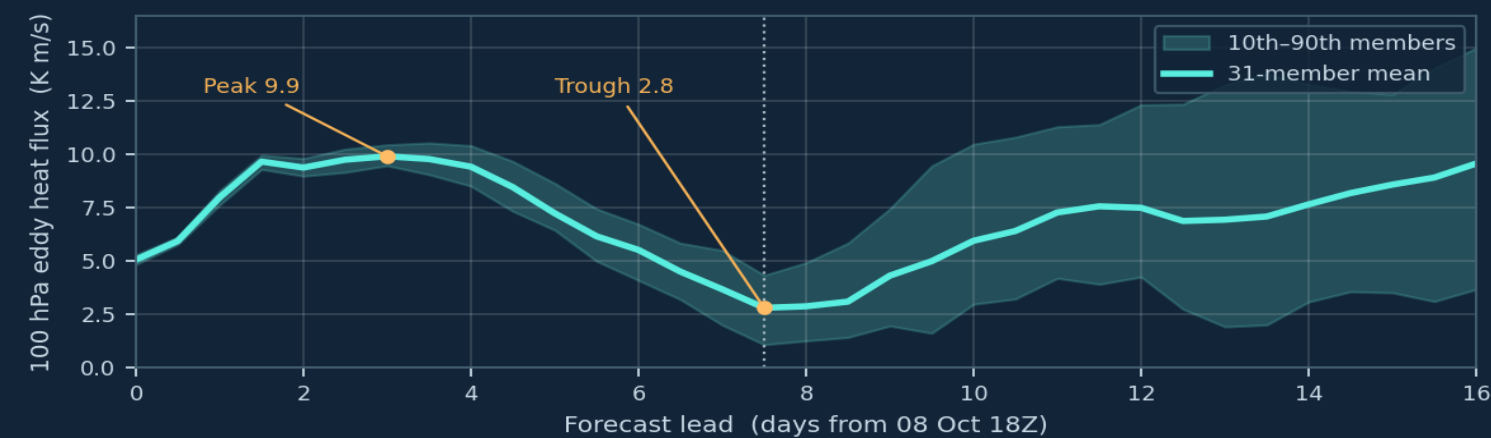
NOAA PSL publishes operational daily and 7-day 500 hPa analyses through 08 Oct. Those gridded fields were not retrieved as a verified 02-08 Oct sequence; do not treat these GEFS forecast maps as the last seven days of observed H500 evolution.

STRATOSCOPE / PLANETARY-WAVE DIAGNOSTICS

Evidence-led assessment | GEFS 08 OCT 18Z | heat flux and vortex forecasts | no direct breaking diagnosis

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100 hPa POLEWARD EDDY HEAT FLUX / 45-75°N



THREE CORE FINDINGS

01 Early upward-wave-driving signal:

Mean heat flux rises from +5.1 at +0h to +9.9 K m/s at +72h (11 Oct 18Z).

02 Weaker forcing follows:

The mean declines to +2.8 K m/s by +180h (16 Oct 06Z), then recovers later.

03 No forecast vortex reversal:

All 133 members in GEFS / IFS ENS / AIFS ENS remain westerly at 60°N, 10 hPa.

ROSSBY-WAVE BREAKING / DIAGNOSTIC SCREEN

500 hPa trough-ridge amplification does not itself confirm Rossby-wave breaking. In the 12 Oct forecast, the North Atlantic sector is strongly negative (-160 m) while Europe is positive (+58 m); this is a baroclinic/planetary-wave pattern worth tracking, not a diagnosed overturning event.

A defensible breaking diagnosis requires a dynamical-tropopause PV map (e.g. the 2-PVU contour on 320-350 K surfaces), irreversible PV-contour overturning/gradient reversal, jet geometry, and consistent evolution in successive valid times. Anticyclonic versus cyclonic breaking cannot be classified from the current Stratoscope diagnostics.

FORECAST TRANSLATION

D0-3: positive but uncalibrated wave forcing; high confidence in westerly 10 hPa flow. D4-10: North Atlantic trough / European ridge contrast; no proven RWB or downward response. D11-16: spread grows; wave forcing recovers in GEFS while vortex remains westerly. Confidence in tropospheric coupling: low / unverified.